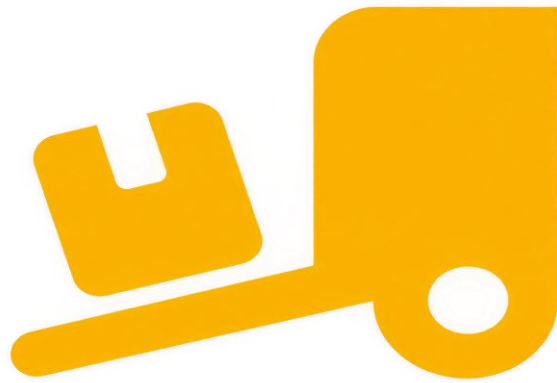




LogiTrack

Logistics & Delivery Management System

1.1 TITLE OF THE PROJECT

LOGITRACK – LOGISTICS & DELIVERY MANAGEMENT SYSTEM

1.2 INTRODUCTION

LogiTrack is a web-based Logistics and Delivery Management System developed to simplify and automate the shipment and parcel delivery process. The system helps customers book shipments online, track their parcels in real time, and receive delivery updates without visiting logistics offices.

The application is developed using **Python (Flask)** as the backend framework and **HTML, CSS, Bootstrap, and JavaScript** for the frontend. **MySQL** is used as the database to securely store customer, shipment, driver, and payment information.

The system consists of three major user roles: **Admin, Customer, and Driver**. Customers can register, log in, book shipments, track parcel status, and download invoices. Admins manage customers, shipments, drivers, payments, and generate reports. Drivers can view assigned deliveries, update shipment status, upload delivery proof, and maintain delivery history.

The application also includes features like shipment tracking, shipping cost calculation, dashboard analytics, PDF invoice generation, and secure user authentication. LogiTrack improves operational efficiency, reduces manual paperwork, minimizes delivery errors, and provides a faster and more reliable logistics management solution.

1.3 OBJECTIVE

- To develop a web-based logistics and delivery management system.
- To provide secure registration and login for Admin, Customer, and Driver.
- To enable customers to book shipments online.
- To generate unique tracking numbers for every shipment.
- To provide real-time shipment tracking.
- To simplify driver assignment by the administrator.
- To calculate shipping costs automatically.
- To generate invoices for completed shipments.
- To improve logistics management through digital records.
- To reduce manual work and improve delivery efficiency.

1.4 SCOPE

The scope of LogiTrack includes the complete automation of logistics operations from shipment booking to final delivery confirmation. The system provides a centralized platform where customers, administrators, and delivery drivers can efficiently perform their respective tasks.

The project includes:

- Customer Registration and Login
- Online Shipment Booking
- Shipment Tracking using Tracking Number
- Driver Management
- Shipment Assignment
- Delivery Status Updates
- Payment Management
- Invoice Generation
- Dashboard Reports
- Analytics and Performance Monitoring

Future enhancements such as GPS tracking, QR code scanning, SMS notifications, and mobile applications can also be integrated.

1.5 HARDWARE AND SOFTWARE INTERFACE

1.5.1 HARDWARE REQUIREMENTS

- Processor : Intel Core i3 or Higher
- RAM : 4 GB Minimum (8 GB Recommended)
- Storage : 500 MB Free Space
- Internet Connection : Required
- Printer : Optional (Invoice Printing)

1.5.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 10/11
- Backend : Python 3.x
- Framework : Flask
- Database : MySQL
- Frontend : HTML5, CSS3, Bootstrap 5, JavaScript
- ORM : SQLAlchemy
- Authentication : Flask-Login
- IDE : Visual Studio Code
- Browser : Google Chrome / Microsoft Edge

1.6 PROGRAMMING LANGUAGES

- **Python** – Backend application development
- **HTML5** – Structure of web pages
- **CSS3** – User Interface Design
- **Bootstrap 5** – Responsive Web Design
- **JavaScript** – Client-side Validation and Interactivity
- **SQL (MySQL)** – Database Management
- **Flask Framework** – Web Application Development

1.7 PROJECT MODULES

ADMIN MODULE

The admin module manages the entire logistics system. It allows the administrator to manage customers, drivers, shipments, branches, and payments. The admin can assign drivers, update shipment status, monitor deliveries, and generate reports for efficient logistics management.

CUSTOMER MODULE

The Customer module allows users to register, log in, and book shipments online. Customers can enter pickup and delivery details, calculate shipping costs, track their shipments using a tracking number, view booking history, download invoices, and update their profile.

DRIVER MODULE

The Driver module enables drivers to log in and view assigned deliveries. Drivers can update shipment status, upload delivery proof, mark shipments as delivered, and view their delivery history.

SHIPMENT MANAGEMENT

The Shipment Management module controls the complete delivery process. It generates tracking numbers, stores shipment details, assigns drivers, updates delivery status, and generates invoices after successful delivery.

1.8 FUTURE SCOPE

- Google Maps integration for pickup and delivery locations.
- QR Code-based parcel tracking.
- Email notifications for shipment updates.
- SMS alerts using Twilio.
- Mobile application for Android and iOS.
- AI-based delivery route optimization.
- Online payment gateway integration.
- Export reports to Excel and PDF.
- Dark mode interface.
- Driver performance analytics.
- Customer feedback and rating system.